
🧪 Amazon CloudWatch – Hands-On Guide

🔧 Prerequisites:

- AWS Free Tier account
- One running **Amazon EC2 instance** (Amazon Linux 2)
- IAM permissions: `CloudWatchFullAccess`, `AmazonEC2ReadOnlyAccess`

📘 1. Monitor an EC2 Instance

✅ Step 1: Enable EC2 Detailed Monitoring (optional)

1. Go to **EC2 Dashboard**
2. Select your instance → Click **Actions > Monitor and troubleshoot > Manage detailed monitoring**
3. Enable **1-minute monitoring** (note: this incurs cost)

✅ Step 2: Install CloudWatch Agent (for memory/disk metrics)

SSH into your EC2 and run:

```
sudo yum install amazon-cloudwatch-agent -y
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
```

During config:

- Collect CPU, memory, disk

- Output to CloudWatch
- Save the JSON config

Then start the agent:

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
  -a fetch-config -m ec2 \
  -c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json -s
```

Check metrics in CloudWatch > Metrics > `CWAgent`.

📊 2. Create a CloudWatch Alarm

✅ Goal: Trigger an alarm if CPU > 70% for 5 mins

1. Go to **CloudWatch > Alarms > Create Alarm**
2. **Select metric** → Browse: `EC2 > Per-Instance Metrics > CPUUtilization`
3. Choose the instance metric
4. Set threshold:
   - Condition: `Greater than 70`
   - Evaluation: `5 minutes`
5. Notification:
   - Choose or create an **SNS topic** (email alerts)
   - Confirm email subscription
6. Name your alarm and create it

🎯 *Test it:* Run CPU stress on EC2 (for demo purposes only):

sudo yum install stress -y
stress --cpu 2 --timeout 600

📈 3. Create a CloudWatch Dashboard

✅ Goal: Visualize EC2 CPU, Memory, and Alarm status

1. Go to **CloudWatch > Dashboards > Create dashboard**
2. Name it: `MyInfraDashboard`
3. Add **Widgets**:
   - **Line graph** for CPUUtilization
   - **Number** widget for MemoryUtilization (from CWAgent)
   - **Alarm status** widget for active alarms
4. Arrange and save the dashboard

🎯 Optional:

- Add widgets from other services (Lambda, RDS, S3)
- Share dashboard via IAM or cross-account

🧠 Bonus: Event-Driven Automation with CloudWatch

- Create a CloudWatch **Alarm Action** → trigger **Lambda** to reboot an instance when a metric exceeds threshold
- Use **CloudWatch Logs Insights** to analyze logs:

fields @timestamp, @message
| sort @timestamp desc

| filter @message like /ERROR/
| limit 20

✅ Outcome

- You now monitor key metrics on EC2 using CloudWatch
- Created alarms to stay proactive
- Built a dashboard for real-time observability

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